

TCP / IP SUITE

1. Create a side-by-side comparison table showing at least 4 key differences between the OSI Model and the TCP/IP Model, including the number of layers, layer names, and where protocols like HTTP, TCP, and IP fit.

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Comparison	OSI Model	TCP/IP Model
Number of layers	7 layers	4 layers
Layer names	Physical, Data Link, Network, Transport, Session, Presentation, Application	Network Access, Internet, Transport, Application
Application protocols	Application Layer (Layer 7): HTTP, FTP, DNS	Application Layer: HTTP, FTP, DNS
Transport protocols	Transport Layer (Layer 4): TCP, UDP	Transport Layer: TCP, UDP
IP protocol	Network Layer (Layer 3): IP	Internet Layer: IP
Ethernet	Mainly Data Link (Layer 2) and Physical (Layer 1)	Network Access Layer
Session & Presentation	Separate Session (5) and Presentation (6) layers	Functions are generally handled within the Application layer
Main purpose	A detailed reference/conceptual model for understanding networking	A practical protocol model used for Internet communication

2. Open Wireshark, start a capture on your active network interface, and visit any website (like www.youtube.com) in your browser. Stop the capture and find a TCP packet in the list. Write down the source IP, destination IP, source port, destination port, and the TCP flag(s) set for that packet.

-> I can't access or control Wireshark on your computer, so I can't truthfully provide values from your capture. You can get the required information in under a minute:

1. Open **Wireshark** and select your active **Wi-Fi/Ethernet** interface.
2. Click **Start Capture**.
3. Open your browser and visit www.youtube.com.
4. Return to Wireshark and click **Stop**.
5. In the display filter, enter:

tcp

6. Select any TCP packet from the packet list.
7. In the **Packet Details** pane, expand **Transmission Control Protocol** and **Internet Protocol Version 4/6**.

Record these fields:

Required information	Where to find it
Source IP	Internet Protocol → Source
Destination IP	Internet Protocol → Destination
Source Port	TCP → Source Port
Destination Port	TCP → Destination Port
TCP Flags	TCP → Flags

Example format

Source IP: 192.168.1.10

Destination IP: 142.250.x.x

Source Port: 52341

Destination Port: 443

TCP Flags: SYN

3. In Wireshark, filter packets by 'udp' and find a DNS query. Identify and note which TCP/IP layer each of the following fields belongs to: source IP, destination IP, source port, destination port, DNS query name.

-> **DNS Query in Wireshark**

First, enter this filter in Wireshark:

udp

Then look for a packet containing **Domain Name System (DNS)**. Expand **Internet Protocol, User Datagram Protocol, and Domain Name System** in the Packet Details pane.

Field	Example	TCP/IP Layer
Source IP	192.168.1.10	Internet Layer
Destination IP	8.8.8.8	Internet Layer
Source Port	54321	Transport Layer
Destination Port	53	Transport Layer
DNS Query Name	www.example.com	Application Layer

Layer breakdown

Application Layer

└─ DNS Query Name → www.example.com

Transport Layer

└─ Source Port → 54321

└─ Destination Port → 53

Internet Layer

└─ Source IP → 192.168.1.10

└─ Destination IP → 8.8.8.8

Important: DNS normally uses **UDP port 53** for traditional DNS queries. The IP addresses and source port will depend on your own computer and capture, so use the actual values shown in your Wireshark packet.

4. Analyze any captured IPv6 packet in Wireshark and list the differences you observe in the IP header compared to an IPv4 packet.

Hint: Look for differences in address format, header length, and any extra fields present in IPv6.

-> **IPv4 vs IPv6 Header Analysis in Wireshark**

In Wireshark, select an IPv6 packet and expand **Internet Protocol Version 6**. Compare it with an IPv4 packet by expanding **Internet Protocol Version 4**.

Feature	IPv4	IPv6
Address format	32-bit, e.g. 192.168.1.10	128-bit, e.g. 2001:db8::1
Header length	Variable: 20–60 bytes	Fixed: 40 bytes
Address size	4 bytes per address	16 bytes per address
Header checksum	Present	Not present
TTL / Hop Limit	TTL field	Hop Limit field
Protocol / Next Header	Protocol field identifies TCP, UDP, etc.	Next Header field performs this function
Fragmentation	Fragmentation information is in the main IPv4 header	Fragmentation uses an optional Fragment Extension Header
Options / extensions	IPv4 options can be included in the main header	Optional information is handled through Extension Headers
Flow identification	No equivalent Flow Label	Has a Flow Label field

What you should observe in Wireshark

For an IPv6 packet, you'll typically see:

Internet Protocol Version 6

Version: 6

Traffic Class: ...

Flow Label: ...

Payload Length: ...

Next Header: TCP/UDP/...

Hop Limit: ...

Source Address: 2001:...

Destination Address: 2001:...

